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Performance test and finite element modeling of variable damping viscous damper

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ABSTRACT

The application of traditional fluid viscous dampers was constrained by their typically constant damping coefficient, which produced a narrow range of damping force. To address this, this paper develops a novel variable damping viscous damper (VDVD) with the feature of variable damping coefficient and force in the passive control domain. The performance tests of VDVD under various loading velocities were carried out to investigate the variable damping characteristic. Simultaneously, tests were conducted to investigate the variable damping characteristics of two critical design parameters: the pre-pressure force of the spring and the shape of the damping orifice. Furthermore, the finite element (FE) model of the device was built using computational fluid dynamics (CFD) method, and verification and parameter analysis were conducted. Through experimental study and finite element simulations, the design philosophy of VDVD was verified. Both experimental and FE analysis show that the shape of damping orifice and pre-pressure of spring will significantly affect the process and critical velocity of variable damping respectively. The residual area and shape of the damping orifice impact the magnitude of the damping force directly. This study offers a thorough comprehension for the design of VDVD which will promote the development of variable damping devices.

1. Introduction

Since the concept of vibration control was introduced in 1960, the field of vibration control has undergone profound development. Passive control [1–3], exhibits extensive utility owing to its inherent simplicity and independent of external energy sources, for instance, fluid viscous dampers [4,5]. However, it is essential to acknowledge that fluid viscous dampers are constrained by fixed damping coefficients and a limited range of output forces, thereby presenting limitations on their applicability, especially in seismic events characterized by a broad frequency range.

In order to overcome this drawback and enhance the efficiency of structural vibration control, scholars have delved into the realm of variable damping. Active control [6–8], while effective, is complex and costly, demanding high-speed, large-capacity actuators and substantial power energy. Semi-active control, proposed by Hrovat in 1985 [9], positioned as a compromise between passive and active control, requires minimal external energy while demonstrating adaptability to structural responses. Regarding the variable damping technique in semi-active control, researchers have proposed varieties of dampers and devices to achieve it. Shi et al. [10] proposed a novel sustainable adaptive passive eddy current-tuned mass damper with variable damping and upgraded it to a semi-active system. By adjusting the air gap, the eddy current damping can be adjusted in real-time. Li et al. [11] proposed a new type of magnetorheological elastomer (MRE) adaptive vibration isolator, which consists of steel and MRE layers. By altering the electric current, the magnetic field is modified, leading to a variation in the damping ratio within the range of 7.8 %–11.2 %. Yu et al. [12] introduced a novel compact rotary magneto-rheological (MR) damper with variable damping, it tunes the yield stress of MR fluid in both pre-yield and post-yield to produce variable damping. Kurino et al. [13] embedded a controller in the viscous damper to achieve variable damping by changing the number of damping orifices. Pinkaew et al. [14] designed a semi-active tuned mass damper (STMD) with variable damping using an optimal control algorithm, which achieve same control performance with ¼ mass of traditional TMD. Sun et al. [15] proposed that STMD has a variable damping ratio by a new short-time Fourier transform (STFT)-based control algorithm. Cundumi et al. [16] proposed a variable

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damping semi-active system, which compose of two fixed-orifice viscous fluid dampers and a variable damping semi-active damper. The two viscous dampers and the variable damping damper are installed in a Y-form arrangement. The damping characteristics are varied by algorithmically altering the position of the variable damping damper. Despite the fact that the variable damping technique achieved by semi-active control can offer real-time adjustable damping force and feedback control, it relies on an extensive sensor network, thereby amplifying concerns about its complexity and reliability in extreme conditions.

Owning to the dependability inherent in passive control, numerous scholars have made efforts to integrate passive control with the concept of variable damping. One way to achieve variable damping is to change the liquid properties of a viscous damper, such as shear-thickening fluid [17–19]. Shear-thickening fluid with different shear rates can generate different liquid viscosities, thus producing different damping. However, temperature-sensitivity constrains its extensive utilization. Another way to achieve variable damping is to change the parameter settings of the damper. Jia et al. [20] designed a damper with a variable annular gap. The damper cylinder has the maximum cross-sectional radius at its midpoint, gradually decreasing from the center towards both ends. The gap becomes smaller as the displacement of the piston increases, which makes the damping coefficient increase correspondingly. Xu et al. [21] designed a new viscous damper with variable damping coefficient by changing the inner diameter which is based on the conventional annular gap-type viscous fluid damper (VFD). Murakami et al. [22] proposed a passive magneto-rheological (MR), in which damping force has a strong dependence on the displacement.

The above studies aim to mechanically increase or vary damping force, but the variable damping devices are related to the loading displacement. Some researchers have also investigated velocity-related variable damping systems. Javadinasab et al. [23] proposed an innovative velocity-dependent adaptive viscous damper (AVD), comprising a twin cone and a ring that are interconnected by eight springs. The area of orifice can be adjusted using a straightforward strategy, where the damping coefficient of the AVD can be set to any value within the range of 1–8.05 times the passive viscous damping coefficient based on the velocity for effective control. Nevertheless, the damping orifice remains located within the cylinder. Once the size of damping orifice is established, its variable damping range is also established, which complicates the process of adjustment and replacement. A real-time, velocity-related and damping orifice shape and size can be easily changed, achieving continuous conversion from CDVD to VDVD passive variable damping device has yet to be introduced based on the limited understanding of the author.

This research aims to design and test a novel variable damping damper. The organization of this paper is as follows: Section 2 provides a brief introduction to the design and working principles of the VDVD. In section 3, the performance of the device is evaluated through experimental tests. Sections 4 and 5 present a simplified finite element model of the device and parametric analysis of critical design parameters respectively.

2. Construction and working principle of VDVD

An innovative variable damping viscous damper (VDVD) is introduced in this study, an enhancement of the traditional constant damping viscous damper (CDVD). The area of the damping orifice is changed by the movement of the movable disk, resulting in a variation in damping coefficient.

2.1. The design philosophy of the VDVD

When structures are subject to high-intensity hazards, a high damping coefficient is demanded by structure to dissipate more energy and protect structural components from damage. Thus, it is necessary to design a damping device for achieving a high damping coefficient when structures undergo extreme conditions.

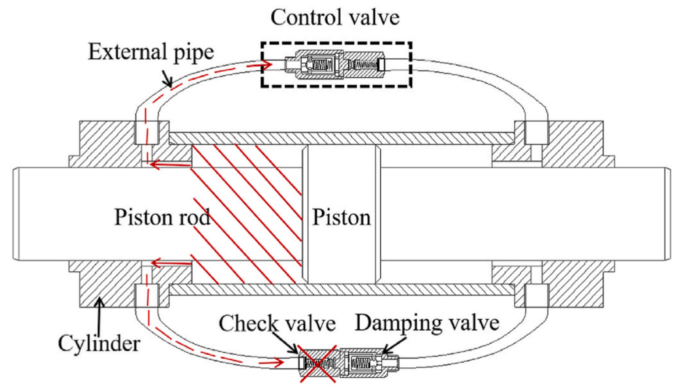

Fig. 1. Design of variable damping viscous damper.

Ou et al. [24] derives the theoretical formulas of damping force and damping coefficient for the circle orifices viscous fluid damper:

$$F = C \cdot v^m \tag{1}$$

$$C = \left(\frac{3m+1}{mn}\right)^m \cdot \left(D^2 - d^2\right)^m \cdot r^{(-3m-1)} \cdot \pi k L \left(D^2 - d^2\right) \tag{2}$$

Where C is the damping coefficient; D is the inner diameter of the cylinder; d is diameter of piston rod; r is the radius of the circular orifice; k is the consistency coefficient; m is the viscosity of the fluid; L is the length of the circular orifice; n is the number of circular orifices; v is the speed of the piston.

The Newtonian fluid ($m = 1$) is most used in viscous fluid dampers. As such, the damping coefficient C can be rewritten as:

$$C = \frac{\alpha}{r^{(3m+1)}} \xrightarrow{m=1} C = \frac{\alpha}{r^4} \tag{3}$$

where $\alpha = \left(\frac{3m+1}{mn}\right)^m \left(D^2 - d^2\right)^m \pi k L \left(D^2 - d^2\right)$ which is determined by the geometry of the damper.

It can be seen from equation (3) that the damping coefficient is inversely proportional to the fourth power of the orifice radius (i.e. the square of area for circle orifice). In this regard, the damping coefficient, as well as damping force, will increase significantly with the area of damping orifice diminishes. Therefore, it is possible to change the damping orifice area dynamically to produce an adequate damping coefficient and damping force for various conditions.

2.2. The configuration of the VDVD

The VDVD consists of a damping cylinder, two control valves, external pipes, etc., as shown in Fig. 1. The damping cylinder is composed of a cylinder, piston, and piston rod. The control valve consists of a damping valve and a check valve. As depicted in Fig. 2. The damping valve is composed of the fixed disk and movable disk, spring, valve body and support nut. The piston forms a perfect seal with the cylinder, and the damping oil can only flow between the control valves on both sides through the external pipes. The damping valve is connected to the check valve in series, ensuring unidirectional flow of damping oil from the former to the latter. There will be pressure differences between the left and right sides of the inflow orifice when damping oil flows through it. The pressure difference will result in the movement of the movable disk to the left and continually decrease the area of damping orifices, which increases the damping coefficient.

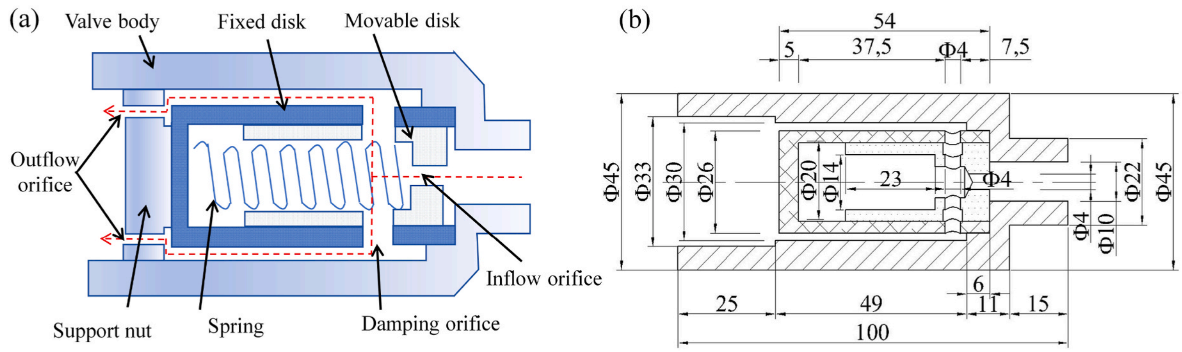

Fig. 2. Schematic and dimension of damping valve: (a) schematic; (b) dimension.

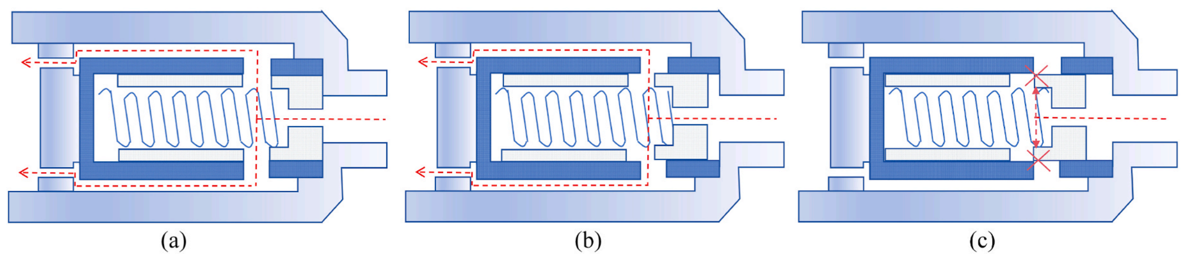

Fig. 3. The different stages of damping valve: (a) initial state; (b) semi-closed state; (c) fully closed state.

2.3. *Working principle of VDVD*

(a) The principle of VDVD

As shown in Fig. 1, when the piston rod moves to the left, the damping oil on the left side is compressed and flows into the external pipes on both sides. The control valves on either side of the damper cylinder are positioned in an asymmetrical manner. The oil in the upper pipes enters the cylinder from the left side chamber through the control valve to the right side chamber, while the damping oil in the lower pipes is unable to pass through the control valve as the control valve is placed reversely and the check value stop the flow of oil. The movement of the piston rod to the right initiates a similar sequence, but the oil can only flow through the control valve on the lower side, so that the characteristic of variable damping can be achieved at both movement directions of the piston rod.

(b) The principle of damping valve

Fig. 3 illustrates the three typical working statues of damping valves. For spring with pre-pressure force, when the pressure difference between the left and right sides of inflow is lower than the pre-pressure force of spring (as shown in Fig. 3 (a)), the movable and fixed disks maintain a full alignment about the damping orifice, which corresponds to the maximum orifice area. Once the pressure difference is greater than the pre-pressure of spring, the movable disk starts to move relative to the fixed disk (Fig. 3 (b)), causing a reduction in the damping orifice area. Based on equation (3), the deduction in the damping orifice area raises the damping coefficient and damping force. In extreme conditions, as shown in Fig. 3 (c), there is a chance that the damping orifices

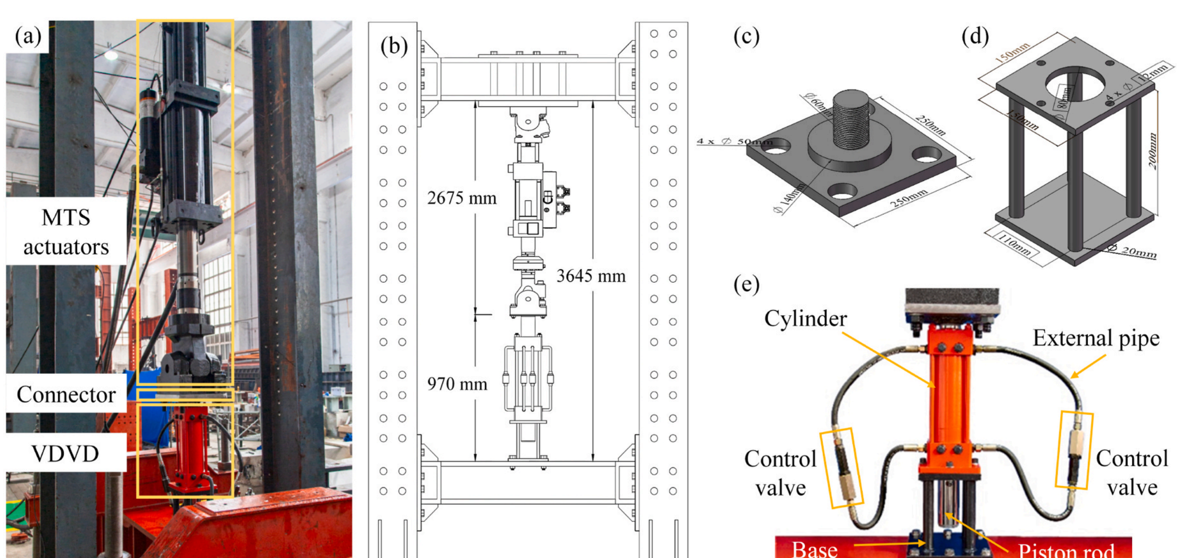

Fig. 4. Test setup and VDVD: (a) test setup; (b) dimension of test setup; (c) connector between VDVD and actuators; (d) base of VDVD; (e) photo of VDVD.

Table 1
Loading protocol.

Case	Frequency/Hz	Magnitude/mm	Maximum speed/ (mm/s)
1	0.06	±40	12.56
2	0.06	±50	15.70
3	0.06	±60	18.84
4	0.06	±70	21.98
5	0.07	±40	15.07
6	0.07	±50	18.84
7	0.07	±60	22.61
8	0.07	±70	26.38
…	…	…	…
52	0.18	±70	79.13

are closed due to the high pressure difference (i.e., structure undergoes extreme hazard). In such scenarios, VDVD will become a rigid body to provide sufficient reaction force for structure. Alternatively, a constant damping coefficient can also be achieved by adjusting the disk length to maintain a minimum orifice.

For cases that spring without pre-pressure force, any pressure difference between two sides of inflow orifice instantaneously decreases the damping orifice area, leading to a rise in damping coefficient. The variation in orifice area is closely related to the pressure difference of inflow orifice, which is velocity-dependent, thus enabling different damping coefficients and forces at different velocities.

3. Performance test of VDVD

This section conducts a series of performance tests of VDVD to evaluate its variable damping properties. Moreover, the effect of the shape of the damping orifice and the pre-pressure force of spring in the damper are also investigated.

3.1. Experimental setup

Performance tests of VDVD were conducted in the structural laboratory of Qingdao University of Technology. The setup of the test system is shown in Fig. 4. The experimental assessments were conducted in a vertical loading configuration, employing a MTS actuator and a VDVD device. These components were interconnected through connectors and situated within a dedicated reaction frame. The MTS actuator featured built-in force and displacement sensors, and data acquisition was facilitated by a high-velocity dynamic system.

The mechanical performance evaluations of the proposed device were conducted in compliance with the standardized protocol outlined in JG/T 209–2012 [25]. The assessment consisted of five repetitive cycles, with the peak damping force being recorded during the third cycle. The sinusoidal displacement loading was applied using the MTS actuator with a maximum capacity of 500 kN. The loading protocol is presented in Table 1. The test is conducted with loading amplitudes of 40 mm, 50

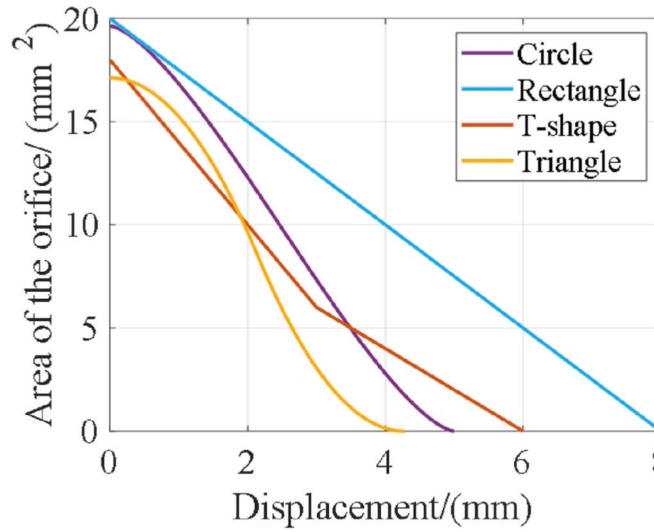

Fig. 6. Residual area of damping orifice with the displacement of movable disk.

mm, 60 mm, 70 mm in a stepwise manner where the loading frequency progressively increases from a lower value to a higher one. The displacement sensors and load cell embedded in the MTS actuator were used in the test to record the time-history data of force and displacement, respectively. The velocity of the loading is the first derivation of the displacement data.

As concluded above, the area of damping orifices is critical to the damping coefficient and damping force. In the proposed device, the residual area is determined by the relative displacement between movable and fixed disks. Under identical relative displacement, diverse orifice geometries result in distinct orifice areas, subsequently altering the rate of change in damping coefficients. Consequently, it is imperative to investigate the effect of various orifice shapes on the performance of VDVD.

For illustration, four damping orifice shapes including circle, triangle, T-shape, and rectangle were selected in the test. The shapes and dimensions of damping orifices are shown in Fig. 5. Fig. 6 illustrates the residual area of damping orifices as the function of the relative displacement of the movable disk. It can be seen that different orifice shapes show different residual areas and descending slopes. In terms of descending slope, the rectangle, T-shape, triangle orifices are equipped with a constant, bi-linear, variable descending rate. As for residual area, the rectangle and triangle orifices have the largest and lowest residual area amount all cases. By comparing the performance of the proposed device with different damping orifice shapes, a comprehensive understanding of damping orifices to device performance can be obtained.

The main design parameters of VDVD and spring are shown in

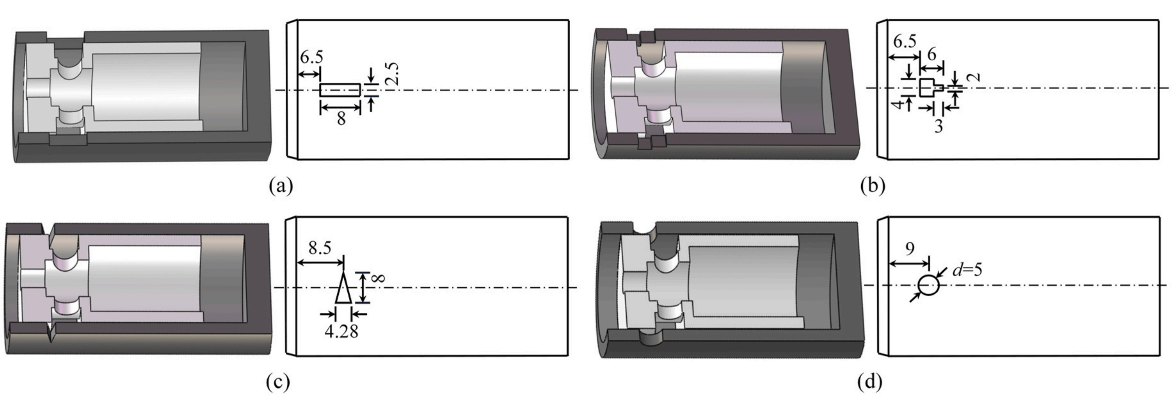

Fig. 5. Damping orifice shapes and dimensions(mm): (a) rectangle orifice; (b) T-shape orifice; (c) triangle orifice; (d) circle orifice.

Table 2
Basic parameters of VDVD.

Cylinder			Piston rod Diameter/mm	Maximum Stroke/mm	Damping oil	
Inside diameter/mm	Outside diameter/mm	Length/mm			Kinematic Viscosity (40 °C)/(mm^2/s)	Viscosity Index
120	140	490	80	±100	45.8	97

Table 3
Information of springs.

Spring type	Spring size (wire diameter × outer diameter × length)/(mm × mm × mm)	Material elastic Modulus/GPa	Spring Stiffness/(N/mm)	Pre-pressure force/N
S-1	1.6 × 12 × 33	190	7.42	0
S-2	1.6 × 12 × 40	190	7.70	53.87

Table 4
Test matrix.

Cases	Spring type	Damping orifice	Loading frequency
1	S-1	T-shape	0.06Hz–0.14Hz
2		Rectangle	
3		Triangle	
4		Circle	
5	S-2	T-shape	0.06Hz–0.18Hz
6		Rectangle	
7		Triangle	
8		Circle	

Table 2 and Table 3. Detailed working conditions and data information are listed in Table 4.

3.2. Results and discussions

3.2.1. Damping valve with different damping orifice shapes

As the movement of movable disk changes the area of damping orifices and damping coefficient, the changing ratio in the area of damping orifices will affect the performance of variable damping. Fig. 7 shows the hysteresis curves of VDVD with different orifice shapes at frequencies of 0.06Hz, 0.08Hz, 0.10Hz and 0.14Hz. In order to better illustrate the difference between various damping orifice shapes, the experimental data was transformed into a velocity versus damping force graph, as shown in Fig. 8. In Fig. 8, the absolute peak damping forces are derived from hysteretic curves in conjunction with their corresponding velocities. The results presented in both Figs. 7 and 8 were obtained using spring S-1 without pre-pressure force in spring.

Fig. 7 reveals that VDVD with different damping orifice shapes displays different values for the maximum damping force and distinct energy dissipation capabilities, even when subjected to the same loading frequency and amplitude. This variation arises from the fact that the residual area of damping orifices of different orifice shapes is diverse under the identical loading frequencies.

Based on the VDVD design philosophy and experimental results, the damping coefficient of VDVD shows a nonlinear relationship with the input velocity. Thus, the power function is adopted to represent the damping coefficient-velocity relationship which is shown in equation (4). The associated damping force as a function of input velocity is given in equation (5). The comparisons between test results and numerical models for different damping orifice are illustrated in Fig. 8, and the corresponding calibration parameters are listed in Table 5.

$$C_v = \begin{cases} c_1 \cdot (|v| - v_c)^p + c_0 & |v| > v_c \\ c_0 & |v| \le v_c \end{cases} \tag{4}$$

$$F_v = \begin{cases} \left\{ c_0 \cdot v_c + [c_1(|v| - v_c)^p + c_0] \cdot (|v| - v_c)^m + F_f \right\} \cdot \mathrm{sgn}(v) & |v| > v_c \\ c_0 \cdot v^m + F_f \mathrm{sgn}(v) & |v| \le v_c \end{cases} \tag{5}$$

Where c_0 is the constant damping coefficient; c_1 is the variable damping parameter; p is the exponent of power function; v is the velocity between the cylinder and piston rod of damper; v_c represents the critical velocity that damping coefficient starts changing which is related to the pre-pressure force of spring; m denotes the viscosity of the fluid; F_f.is the friction force.

It can be seen from Fig. 8 that the numerical model has a good agreement with different damping orifice shapes, which can be applied to the practical design and performance assessment of VDVD. In Fig. 8, significant rises in damping force when the velocity reaches around 30 mm/s across all scenarios. The force-velocity relationship of different orifices is determined by the residual area of damping orifices (Fig. 6) to some extent. As shown in Fig. 6, the residual area of the rectangle orifice is the largest, resulting in the lowest damping force amount all orifice shapes. The T-shape with the second-largest residual area generates the second-to-last damping force. However, the triangle orifice with the smallest residual area does not exhibit the largest damping force. The finite element study on damping orifice shapes proves the correlation between damping orifice residual area and damping force in section 5.1. Thus, the mismatch in damping force and residual area may be caused by the manufacturing defect in the damping valve, leading to an unsmooth movement of the movable disk.

3.2.2. Pre-pressure force of spring

In the damping valve, when the length of the spring exceeds the inner space between the movable disk and the fixed disk, the spring will be pre-compressed, resulting in the pre-pressure force in spring. In this part, the damping valve with and without pre-pressure force of spring was used in the test to investigate its effect on the performance of VDVD. Fig. 9 displays the hysteresis curves for orifice shapes with and without pre-pressure at frequencies of 0.08Hz, 0.10Hz, 0.12Hz and 0.14Hz. Fig. 10 shows the comparisons of test and numerical model results in terms of the maximum damping force-velocity relationship (with pre-pressure). Table 6 shows the range of damping forces generated at corresponding velocities in the VDVD test.

Fig. 9 illustrates the effect of pre-pressure on the damping force. It can be observed that each damping orifice shape(Fig. 9 (a)–(d)), damping force without pre-pressure (0.08Hz WOP, 0.10Hz WOP, 0.12Hz WOP, 0.14Hz WOP) increase significantly, while those cases with pre-pressure (0.08Hz WP, 0.10Hz WP, 0.12Hz WP, 0.14Hz WP) behaves similarly to CDVD under same loading frequency. The reason is that when there is pre-pressure force, the loading velocity is less than a specific threshold (liquid pressure at the inflow orifice induced by loading is less than the spring pre-pressure), and the movable disk and the fixed disk are in a relatively static state, so the damping coefficient is a fixed value in this phase, the shape of the hysteresis curve is the same as the CDVD. When the piston velocity increases to a certain threshold, the liquid pressure at the inflow orifice exceeds the spring pre-pressure. As a result, the spring will be compressed, resulting in the movement of the movable disk and decrease in the area of damping orifice, and an increase in both the damping force and damping coefficient.

For test results, Fig. 10 shows a significant difference compared with Fig. 8. At the same velocity, the damping force generated by the VDVD without pre-pressure is significantly higher than that of the VDVD with pre-pressure. This finding is consistent with the data in Fig. 9. In terms of

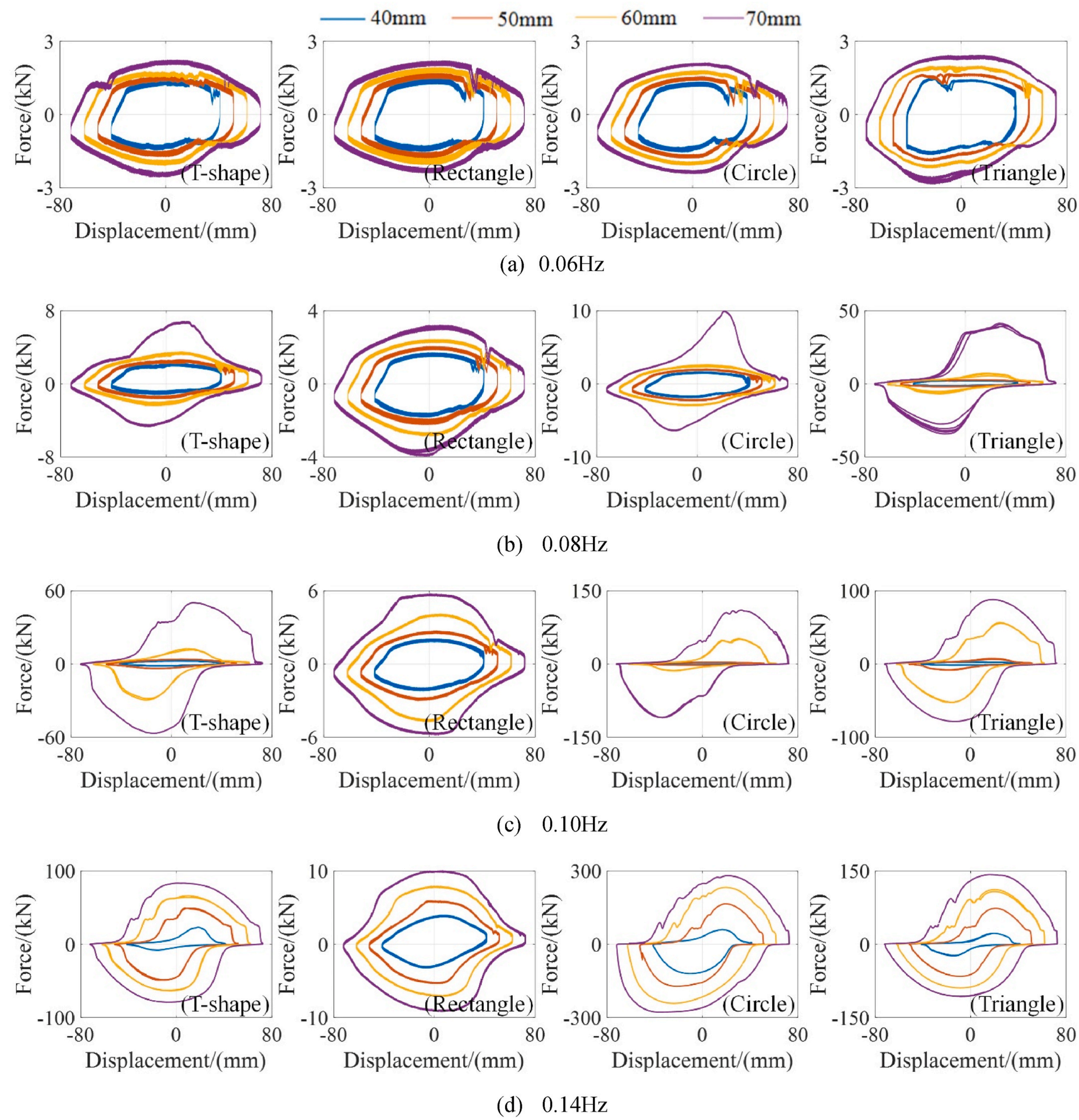

Fig. 7. Displacement -force curves of VDVD with S-1 spring.

numerical model results, the fitted curves are separated into fixed damping segment and variable damping segment at critical velocity in Table 5. The goodness between test and numerical model results for spring with pre-pressure proves the applicability of equations (4) and (5) for different damping orifice shapes and spring pre-pressures.

Table 6 shows the maximum working velocity and damping force for different cases. Due to the capacity of loading equipment, the maximum working velocity and damping force in Table 6 indicate the corresponding value in the tests and may not be the ultimate working values of each testing case. The ultimate capacity of VDVD needs further studies.

4. Finite element model and validation

In this section, the simplified finite element models of VDVD were constructed and verified against the test results.

4.1. Assumptions and simplifications of VDVD for finite element analysis

As the essential component of VDVD, damping valve with a changeable area of damping orifice can generate variable damping force that surpasses the damping force generated by other components such as external pipes and so forth. So, the finite element model of damping

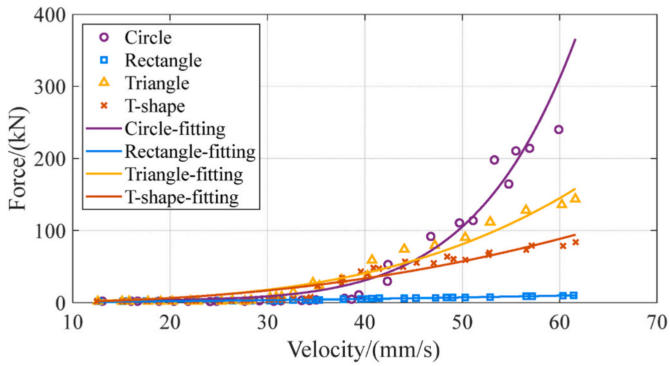

Fig. 8. Test and numerical model results of maximum damping force-velocity relationship (without pre-pressure).

valve was constructed and analyzed in this section. While the damping force generated by the damping pipe, check valve and damping cylinder is calculated using the respective equations provided below.

The damping force generated by the external piping is based on the Darcy equation (6) [26]:

$$h_f = \lambda \frac{lv^2}{2dg} \tag{6}$$

Where h_f is the along-travel resistance loss, λ is the head loss coefficient; l is the pipe length, d is the pipe inner diameter, v is the pipe flow velocity, and g is the acceleration of gravity, λ is head loss coefficient which can be calculated by equation (7):

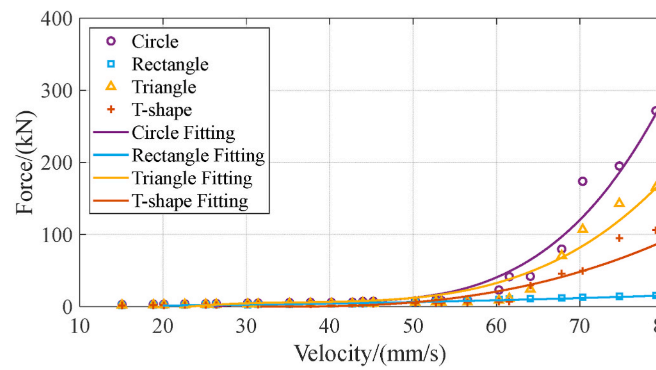

Fig. 10. Velocity-maximum damping force relationship and fitting curves (With pre-pressure).

Table 5
Numerical model calibration parameters.

Case		Critical velocity v_c/(mm/s)	Constant damping coefficient c_0/(kN·s/mm)	Variable damping parameter c_1	Exponent p	Friction force F_f/kN
Without Pressure	Rectangle	0	0.165	0.001	0.502	1.2
	T-shape	0	0.165	0.001	1.757	1.2
	Triangle	0	0.165	0.002	1.699	1.2
	Circle	0	0.165	0.001	1.984	1.2
With Pressure	Rectangle	56.2	0.165	0.001	1.032	1.2
	T-shape	54.4	0.165	0.005	1.566	1.2
	Triangle	53.5	0.165	0.002	1.899	1.2
	Circle	52.6	0.165	0.004	1.964	1.2

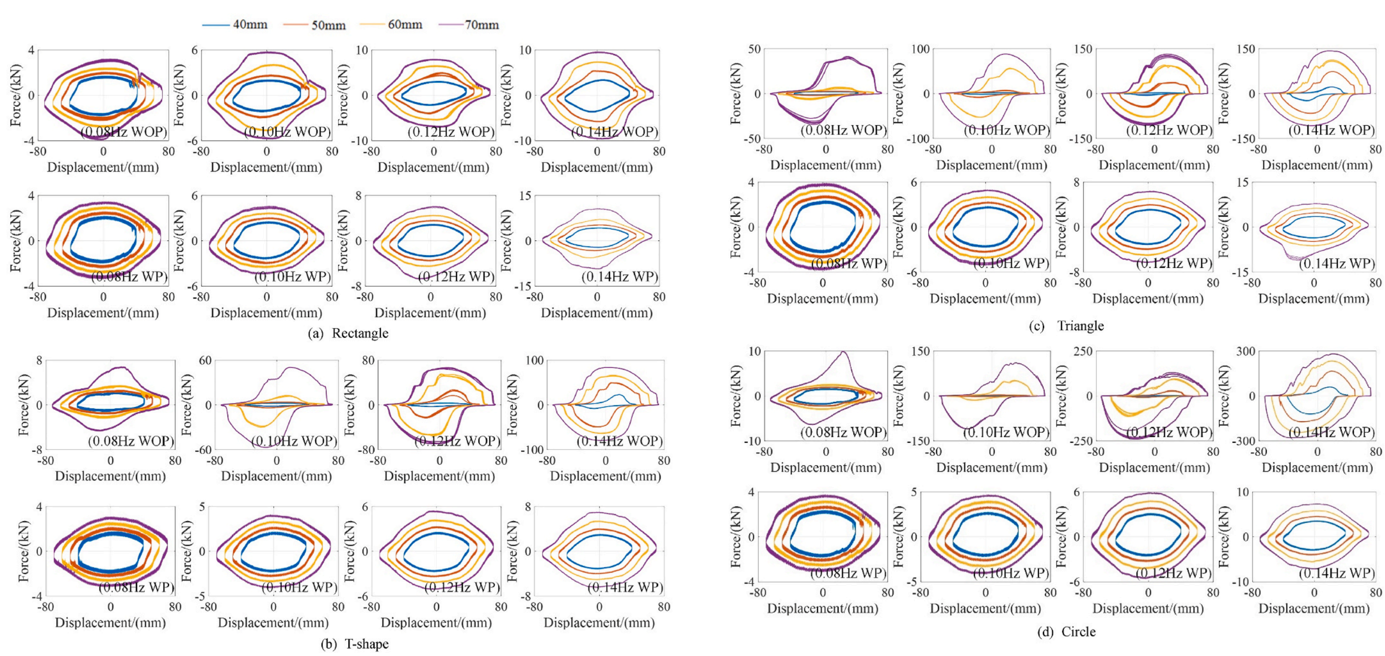

Fig. 9. Hysteresis curves of displacement and force with and without pre-pressure for four orifice shapes (WOP means spring without pre-pressure force; WP means spring with pre-pressure force).

Table 6
Working velocities range and corresponding damping force.

Case	Damping orifice shape	Critical velocity/(mm/s)	Critical force/kN	Maximum velocity/(mm/s)	Maximum force/kN
Without Pre-pressure	Rectangle	–	–	61.54	10.17
	T-shape	–	–	61.54	83.98
	Triangle	–	–	61.54	143.78
	Circle	–	–	61.54	240.11
With Pre-pressure	Rectangle	58.1	2.19	79.13	25.56
	T-shape	60.3	5.36	79.13	79.13
	Triangle	57.5	7.67	79.13	165.33
	Circle	58.8	7.95	79.13	271.54

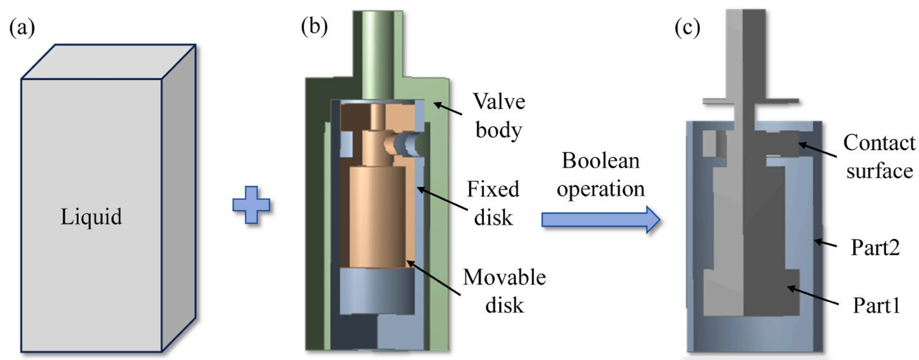

Fig. 11. Boolean operation to create liquid model: (a) cuboid liquid; (b) solid geometric model (c) liquid geometric model.

$$\lambda = \begin{cases} \dfrac{0.00214g}{d^{0.3}} & v > 1.2\mathrm{m/s} \\ \dfrac{0.001824g}{d^{0.3}}\left(1+\dfrac{0.867}{v}\right)^{0.3} & v \leq 1.2\mathrm{m/s} \end{cases} \tag{7}$$

Calculate the pressure loss of the check valve according to the flow relational equation (8) [27]:

$$\Delta P = \rho/2\left(\frac{Q}{C \cdot A}\right)^2 \tag{8}$$

Where Q is the check valve liquid flow; C is the flow coefficient; A is the flow area; ΔP is the pressure loss; ρ is the hydraulic fluid density.

The damping force generated by the damping cylinder is small and is considered negligible in this article, and the damping force of the external piping and check valve is calculated separately.

The overall damping force F is given by the following equation (9), where A_f is the cross-sectional area of the external pipe, A_p is the effective area of the check valve, F_v is the force generated by the damping valve:

$$F = h_f \cdot A_f + \Delta P \cdot A_p + F_v \tag{9}$$

In addition, some assumptions are adopted to simplify the simulation.

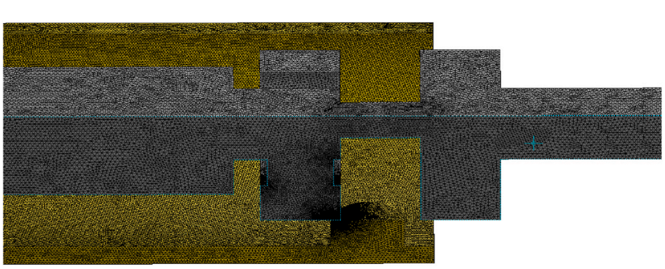

Fig. 12. Damping valve liquid mesh model.

(1) The solid parts of the damping valve, including the valve body, fixed disk and movable disk are considered to be rigid bodies, which means the liquid pressure caused by the deformation of solid parts is neglected.
(2) The spring in the damping valve is simplified as a linear spring force coupled between fixed disk and movable disk. This representation is achieved by UDF macros, which are user-defined macros used to create custom computational models and boundary conditions, allowing for the customization of the simulation process. The deformation of the spring is calculated by the relative displacement between movable disk and fixed disk.
(3) Due to the symmetry of the damping value, a quarter of the model is utilized for computation to simplify the calculation.

With the above assumptions and simplifications, the finite element model is more computationally efficient with consideration of the variable damping coefficient and overall damping force.

4.2. Finite-element model

The damping valve is the core component of the VDVD, which produces the variable damping coefficient. In the damping valve model, since the solid part is assumed to be a rigid body, only the liquid inside the damping valve is modeled in order to simplify the calculation. Liquid and solid models are shown in Fig. 11. The liquid part (Fig. 11 (c)) is generated by the Boolean operation between the liquid cuboid (Fig. 11 (a)) and the solid part (Fig. 11 (b)), where the liquid inside the valve body is retained. A contact surface (the contact area between two parts of liquid in Fig. 11 (c)) was set in the liquid area between the movable disk and the fixed disk, which represents the damping orifice area. When the movable disk moves, liquid will flow from part 1 to part 2, resulting in a modification in the contact surface area. This alteration results in a variation in the size of the damping orifice area. The exterior surface of the liquid in contact with the damping valve (including movable disk, fixed disk and valve body) as a rigid wall boundary condition to reflect the solid-to-fluid effect between damping oil and the value, and the symmetrical dividing surfaces are set to symmetrical boundary

Table 7
Material properties of the model.

	Parameter	Values
Liquid	Properties	Laminar flow
	Density/kg·m^{-3}	880
	Power Viscosity/Pa·s	0.04
Spring	Stiffness/N·mm^{-1}	56.20

Table 8
Simulation configuration table.

Parameters	Value	Parameters	Value
Defaults		Sizing	
Physics Preference	CFD	Growth Rate	1.05
Solver Preference	Fluent	Max Size	3.0e-004 m
Element Order	Linear	Curvature Min Size	2.5e-005 m
Element Size	3.0e-004 m	Curvature Normal Angle	12.0°
Export Format	Standard	Proximity Min Size	1.0e-004 m

conditions.

According to the principle of flow equivalence, the velocity of piston rod movement is converted into the inflow orifice liquid velocity of the damper valve according to equation (10):

$$A_{io} \cdot v_{io} = (A_{dc} - A_{rod}) \cdot v \tag{10}$$

Where A_{io} and v_{io} are the area and flow velocity of the damping valve inflow orifice, respectively; A_{dc} is the inside cross-sectional area of the damping cylinder; A_{rod} is the cross-sectional area of the piston rod; and v is the velocity of the piston rod motion.

The mesh density around damping orifices is increased due to the complex contact relationships. The meshed finite element model is shown in Fig. 12. Table 7 shows the material properties of the liquid and spring. Table 8 gives information about the computational solution method and mesh parameter.

4.3. Model validation

The hysteretic curve of test results and finite element results for cases with different damping orifices and springs without pre-pressure are contrasted and plotted in Fig. 13.

From the comparison results, the finite element results exhibit satisfactory concurrence with the test results. However, it is evident that the hysteretic curves of the test are asymmetrical around the y-axis, while those from the finite element model are symmetrical with the maximum force shown at zero displacement. The disparity arises due to the time-lag in test results. The delay in the test is caused by the friction between the movable disk and the fixed disk. And this is attributed to a manufacturing defect between the movable disk and the fixed disk. The time-lag phenomenon generated in the experiment will be investigated more thoroughly in future studies.

In addition, it is readily apparent to observe the disparity between the maximum and minimum force in test results (Fig. 13 (a) for instance). Due to the manufacturing defect in damping valve, the friction between the fixed disk and the movable disk of one damping value is significantly greater than another, resulting in an increase of equivalent spring pre-pressure force and a delay in the variable damping. These errors can be diminished by improving the manufacturing process.

Table 9 shows the comparison and errors between the finite element simulation results and the test results. The maximum error between the two is 13.71 %. Despite minor disparities between the finite element results and the test results, the difference in maximum force and shape of loops is negligible. This confirms the correctness of the finite element modeling and simulation techniques, rendering them suitable for subsequent simulations.

5. Parameter analysis of VDVD

In this section, the influence of design parameters of damping valve including the shape of damping orifice, stiffness and pre-pressure force

Table 9
Comparison and errors between test and finite element results.

Orifice shapes	Frequency /Hz	Magnitude A/ mm	$\lvert F_{\max,\text{test}}\rvert$/kN	$\lvert F_{\max,\text{FEM}}\rvert$/kN	Error/ %
Circle	0.08	70	9.821	9.688	1.35
	0.09	70	78.351	81.237	3.68
Triangle	0.06	70	2.576	2.265	12.07
	0.07	70	8.217	7.573	7.84
Rectangle	0.13	70	8.774	7.643	12.89
	0.14	70	10.038	9.214	8.21
T-shape	0.08	70	4.587	5.216	13.71
	0.09	70	25.581	24.648	3.65

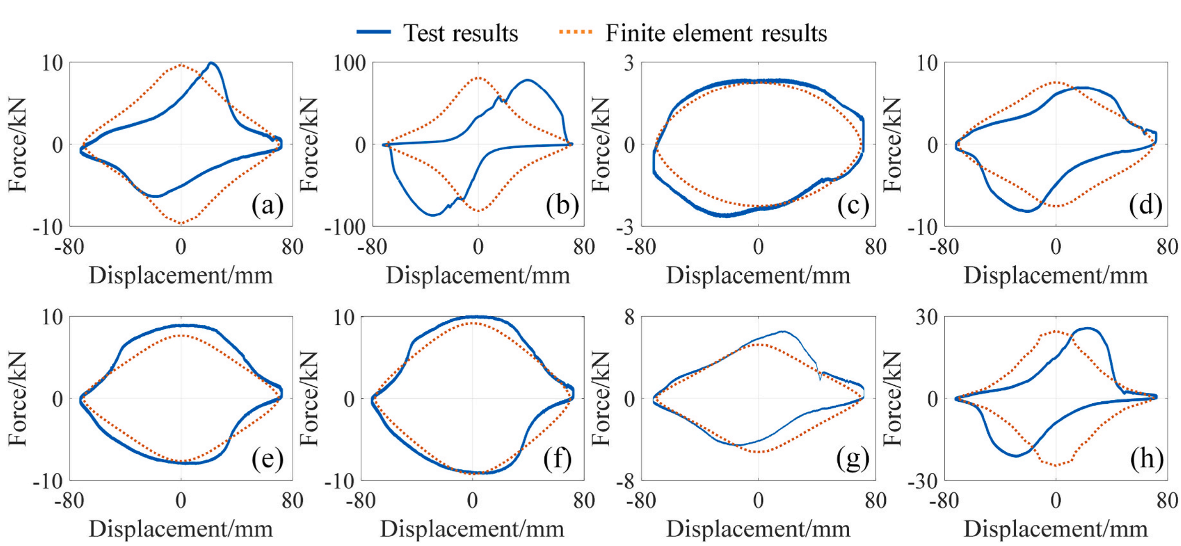

Fig. 13. Hysteresis curves comparison of test results and finite element results:(a)circle with 0.08Hz-70mm; (b) circle with 0.09Hz-70mm; (c) triangle with 0.06Hz-70mm; (d) triangle with 0.07Hz-70mm; (e) rectangle with 0.13Hz-70mm; (f) rectangle with 0.14Hz-70mm; (g) T-shape with 0.08Hz-70mm; (h) T-shape with 0.09Hz-70mm.

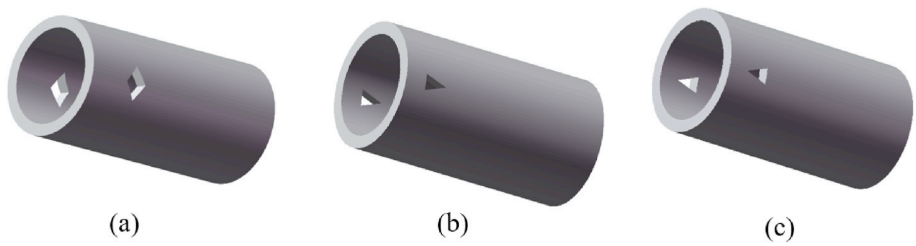

Fig. 14. Damping orifice shapes: (a) rhombus orifice; (b) type-I triangle orifice; (c) type-II triangle orifice.

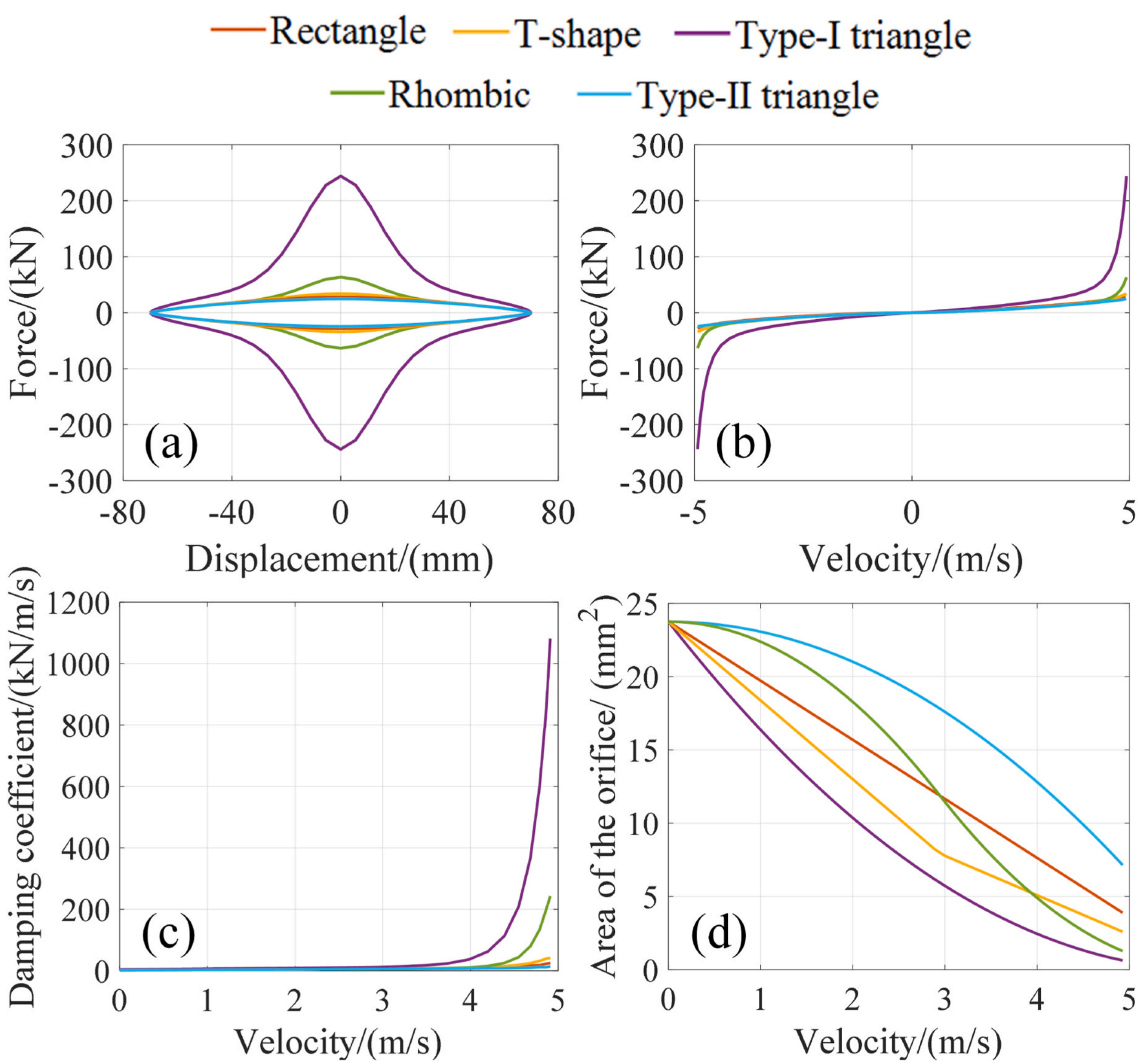

Fig. 15. Analytical curves of different orifice shapes: (a) displacement versus damping force curve; (b) velocity versus damping force curve; (c) damping coefficient curves at different velocities for five orifice shapes; (d) velocity versus damping orifice area curve.

of spring are analyzed and compared.

5.1. Damping orifice shapes

Based on the validated finite element model, some additional damping orifice shapes, including rhombus and two types of triangles, are investigated in this section, as shown in Fig. 14. The orifice shapes all have an initial area of 23.75 mm^2, and the length of all damping orifices is fixed at 5.5 mm. Those identical sets consist of the area and length of damping orifice, which guarantees that the shape of damping orifice is the only factor that will affect the variable damping force. The liquid velocity at inflow orifices is set as a sine wave with a maximum velocity of 4.926 m/s, corresponding to a sine displacement loading with a frequency of 0.14Hz and an amplitude of 70 mm for the damper piston rod.

The comparison results of VDVD with different damping orifices are shown in Fig. 15. Fig. 16 depicts the velocity and pressure diagram at different displacements for the T-shaped orifice.

In Fig. 15 (a), in terms of maximum damping force values, the hysteresis curves show that the maximum force generated by different orifice shapes illustrates distinct differences. The maximum damping

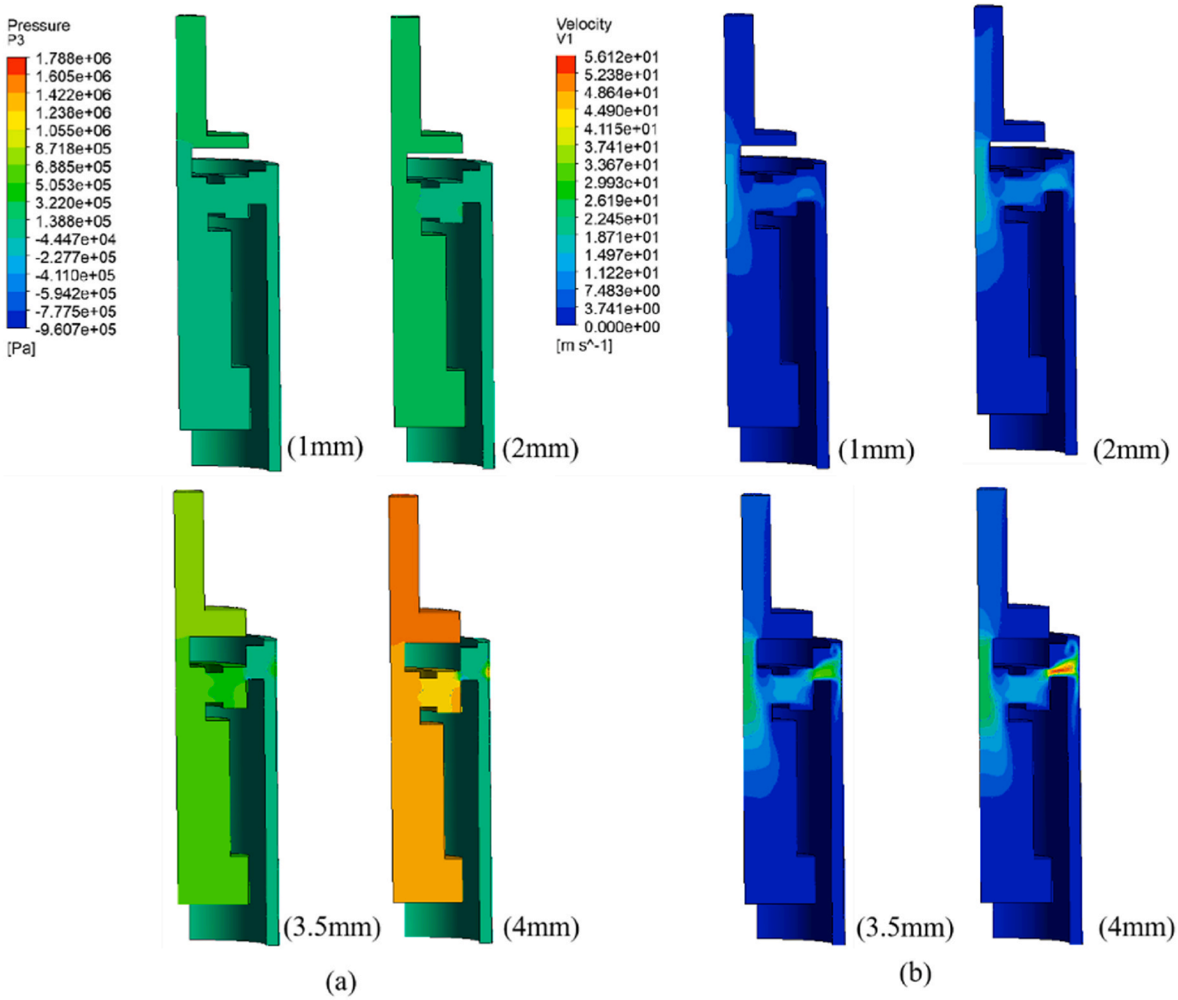

Fig. 16. Liquid pressure and velocity diagram of damping value: (a) Pressure diagram; (b) Velocity diagram (1 mm, 2 mm, 3.5 mm and 4 mm represent the displacement of the movable disk).

force that VDVD with the type-I triangle orifice is about four times greater than that with other shapes. From Fig. 15 (b) velocity versus damping force curve, it is evident that the damping force of VDVD with type-I triangle orifice exhibits the significant increase firstly when the velocity is about 3.98 m/s. Fig. 15 (c) shows that the damping coefficient is relatively small and close for all cases when the input velocity is less than about 3.5 m/s, while the coefficients of the type-I triangle orifice and rhombus orifice shapes are larger compared to other orifice shapes, though the initial area of each case is the same.

Damping orifice area versus velocity for five orifice shapes is shown in Fig. 15 (d), which describes how the residual area of damping orifice varies with input velocity (i.e., movement of the movable disk). At the

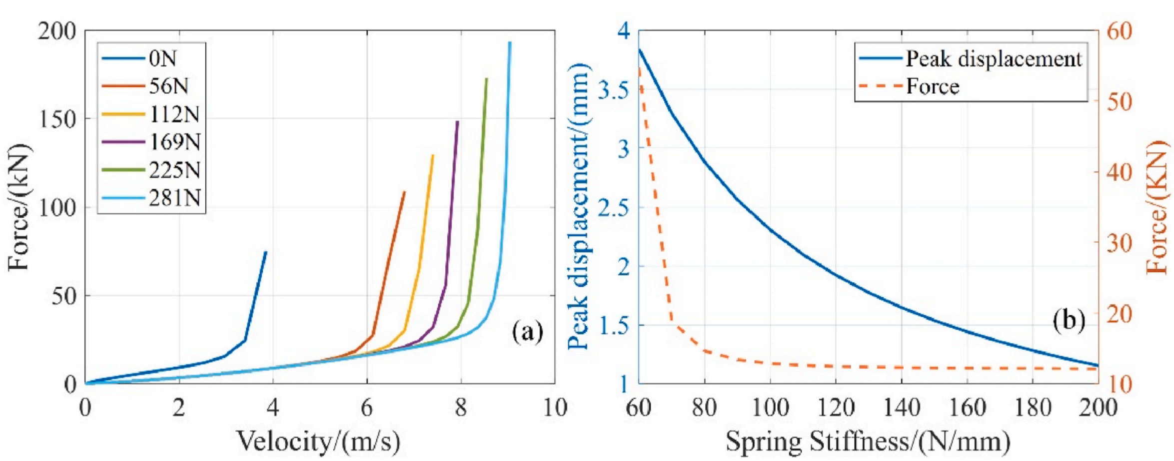

Fig. 17. Spring pre-pressure and stiffness influence curve: (a) velocity versus damping force curves for different spring pre-pressure (0N–281 N corresponds to 0–5 mm of pre-compression); (b) peak movable disk displacement with different spring stiffnesses and corresponding VDVD damping force.

same input velocity, the movement of movable disk for each case is the same, but the residual area of each case is different due to the geometry of damping orifices. For cases like type-I triangle orifice (Fig. 14 (b)), the residual area of damping orifices is the smallest and generates the largest damping force. While, for cases like type-II triangle orifice (Fig. 14 (c)), the residual area is the largest and the corresponding damping force is the smallest. Therefore, the case with the largest covered area (i.e., minimal effective area) of damping orifices will generate the greatest damping force. Overall, the residual area of the damping orifice impacts the magnitude of the damping force directly. Additionally, different shapes of damping orifices are equipped with different area varision properties and result in distinct variations trend in the damping force.

The pressure and velocity diagrams presented in Fig. 16 also illustrate the same conclusion. The pressure difference and velocity at damping orifice rise due to the decrease in damping orifices, finally resulting in a larger damping force and damping coefficient.

5.2. Stiffness and pre-pressure force of spring

The spring stiffness and pre-pressure directly affect the displacement of the movable disk. Fig. 17 presents the influence of spring pre-pressure force and stiffness on the damping force. The simulation utilizes circle damping orifices with a diameter of 5.5 mm in this section.

In Fig. 17 (a), as the pre-pressure increases, the critical velocity that the damping force exhibits nonlinear raises also increases by about 0.5 m/s for every 1-mm increase in the pre-compression of spring. Furthermore, when the stiffness of spring increases (Fig. 17 (b)), the movement of movable disk is also restrained, resulting in a larger damping orifice and a smaller damping force.

Besides, when the displacement of movable disk decreases to a certain value (i.e., stiffness of spring exceeds 80 kN/mm) as shown in Fig. 17 (b), the corresponding damping force does not decrease continuously but keeps a certain value. To conclude, changing the stiffness or pre-pressure of spring can adjust the critical velocity or critical force of variable damping.

5.3. Limitations and suggestions

Based on the test and finite element results, it is evident that the increment of the damping coefficient is most significant when the residual damping orifice area is close to zero. In that case, a small move of the moveable disk (i.e., a small increase in loading velocity) will trigger a high variance in the damping coefficient and force. Thus, a precise control of the residual damping orifice area is critical for different design purposes.

In this regard, it is suggested to use a longer damping orifice with a diminishing size perpendicular to the movement direction of the disk. An ideal orifice shape is similar to the type-I triangle orifice in Fig. 14 (b) but with a larger height. By doing so, the reduction ratio of the damping orifice area goes down for a unit move of the movable disk when the damping orifice area is relatively small. A manageable rise in the damping coefficient is easy to achieve.

6. Conclusions

This paper presents a novel velocity-related variable damping viscous damper (VDVD). The device dynamically adjusts the damping coefficient by modifying the size of the damping orifice through a mechanical setup, enabling real-time variability in the damping coefficient and force output. The design of the device was validated through tests at different velocities. Two critical design parameters were also analyzed during the test. Moreover, the FE model of the VDVD was built and verified against the test results. Subsequently, a series of parametric analyses were performed to investigate the performance-related factors. Several conclusions can be given as follows.

(1) The VDVD achieves real-time variation of the damping coefficient by incorporating a variable damping orifice in the control valve. The movement of movable disk caused by the change in input velocity results in a change of the damping orifices, which enables the velocity-related variable damping coefficient.
(2) Experimental studies validate the design philosophy of the VDVD. Various damping orifice shapes exhibit different damping force hysteretic profiles. The pre-pressure force of spring shows a significant influence on the critical velocity of the variable damping. With an input velocity lower than the critical velocity, the VDVD operates as a traditional CDVD. While it shows the variable damping character once the input velocity exceeds the critical velocity. This unique VDVD design allows smooth conversion between CDVD and VDVD states based on pre-pressure of spring, catering to diverse scenario requirements.
(3) The simplified FE model of VDVD which focuses on the control value was built using the CFD method. Compared with test results, the FE results show a good agreement in both hysteretic results and maximum force within an acceptable error tolerance, which was capable of modeling the behavior of VDVD for parameter analysis.
(4) Based on the parameter analysis of VDVD by FE study, different damping orifice shapes will generate various damping force profiles and peak forces, which can cater to diverse design requirements. Pre-pressure of spring in control value will change the critical velocity of variable damping. With the preset compression of spring, the VDVD is equipped with both constant damping when the input velocity is lower than critical velocity and variable damping when input velocity is greater than critical velocity.

CRediT authorship contribution statement

Mingmei Shi: Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation. **Weiqing Fu:** Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Funding acquisition, Formal analysis. **Mao Li:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Formal analysis. **Haozhe Wang:** Writing – review & editing, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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